MSE 590500 Spring 2026 Mid-term exam

Date: 4/26/2026 (Mon), in-class. Your name:______________

Part I, True or False, circle your choice. 20 points

1. (T) (F) NumPy arrays allow element-wise operations without explicit loops.
2. (T) (F) In Python list, `a = [1, 2]` and `b = [3, 4]`. So, `a + b = [3, 6]`.
3. (T) (F) pandas.DataFrame behaves like numpy array.
4. (T) (F) The expression `b = a` creates a copy of array `a`.
5. (T) (F) Python syntax: `for i in range(5):`, `i` loops from 1 through 5.
6. (T) (F) SymPy performs symbolic (exact) computations rather than numerical approximations.
7. (T) (F) In finite difference methods, decreasing grid spacing generally improves accuracy.
8. (T) (F) SymPy expressions can be converted to numerical functions using lambdify.
9. (T) (F) In a ternary phase diagram, there are three independent variables for the composition.
10. (T) (F) In a binary phase diagram, a tie-line connects two equilibrium phases.

Part II, Single selection, circle your choice. 15 points

1. `a = np.array([1,2,3])`
   What is the result of `b = a * 2`?
   - A. `[1,2,3,1,2,3]`
   - B. `[2,4,6]`
   - C. `[1,4,9]`
   - D. Error
2. Which matplotlib command sets logarithmic y-scale?
   - A. `plt.log()`
   - B. `plt.yscale('log')`
   - C. `plt.semilogx()`
   - D. `plt.scale('ylog')`
3. What does this UNIX command do?
   `cat file.txt | grep "error"`
   - A. Deletes lines containing "error"
   - B. Finds lines containing "error"
   - C. Sorts lines alphabetically
   - D. Counts number of lines
4. In pandas, which function reads a CSV file?
   - A. `pd.load_csv()`
   - B. `pd.read_csv()`

C. `pd.import_csv()`

D. `pd.open_csv()`

5. Which of the following describes a eutectic point?

   A. Single phase region

   B. Two-phase equilibrium

   C. Three-phase equilibrium

   D. Critical point

Part III, Multiple selection, circle your choices. 20 points

1. Which of the following are valid NumPy operations?

   A. `a + b`

   B. `np.dot(a, b)`

   C. `a * b`

   D. `a.append(b)`

2. Which are advantages of using pandas?

   A. Handles tabular data

   B. Supports labeled indexing

   C. Faster than NumPy for all operations

   D. Easy CSV/Excel handling

3. Which statements are correct?

   A. `fig, ax = plt.subplots()` creates both figure and axes

   B. `ax.plot()` plots on a specific axes

   C. `plt.plot()` always refers to the last active axes

   D. Multiple axes can exist in one figure

4. Which statements are correct?

   A. A DataFrame is a 2D labeled data structure

   B. A Series is a 1D labeled array

   C. `df["col"]` selects a column

   D. `df.iloc` uses label-based indexing

5. SymPy vs NumPy

   A. SymPy is used for symbolic math

   B. NumPy is used for numerical computation

   C. SymPy returns exact expressions

   D. NumPy can solve symbolic equations

Part IV, Answer these questions, 25 points

1. Write the command to load `numpy` library. (2 pt)

2. What does the following code do? (3 pt)

```
x = np.linspace(0, 1, 5)
print(x)
```

3. For a regular solution model of Gibbs free energy of mixing: (10 pt)

$$G(x) = \Omega x\big(1 - x\big) + x \ln x + \big(1 - x\big) \ln \big(1 - x\big)$$

Write the code to A) load in required libraries, B) create a composition grid between 0 and 1 with 201 points, C) calculate the corresponding values of Gibbs free energy, D) plot the curve using matplotlib.

4. For $f(x) = \sin(2x)$, write a code to numerically calculate the value of $f'(x_0)$, where $x_0 = 0.375$. (5 pt)

5. Explain how the convex hull method is used to determine phase equilibrium. (5 pt)